Local voice message search tool

"We need a way to search voice messages like we search text. We need a search bar feature for audio. It has to be private. The tool must take your voice files already on your own computer and turn the speech into text. It should tag the timestamp for every word. So when you type "lab requirement " it finds the voice note. Takes you right to the moment that phrase was spoken. The key is that everything must run on your laptop. No uploading your audio no sending your chats to a server ." *Generated by LeapSpace*

Private On-Device Voice Message Search at a Glance

Key Findings Table

Key finding	What it means for your tool	Relevant population / setting	Citations
A fully local voice-search pipeline is technically feasible	You can transcribe local audio, attach timestamps, index results, and search without cloud upload if you assemble local components	On-device ASR, local retrieval, offline system design	[1] [2] [3] [4] [5] [6] [7] [8] [9] [10] [11] [12]
Streaming RNN-T/Conformer-style ASR is the strongest fit for laptop transcription	These architectures repeatedly appear as the practical choice for low-latency, local transcription	On-device / embedded / low-latency speech recognition	[13] [14] [15] [16]
Word-level timestamps are achievable, but timestamp evaluation is inconsistent	Jump-to-moment UX is practical, but exact word timing quality may vary across long or messy recordings	Forced alignment, long-form audio, imperfect transcripts	[5] [6] [17] [18] [19]
Transcript-only search is not robust enough by itself	Exact phrase search should be backed by phonetic/lattice fallback, and optionally semantic retrieval	Spoken term retrieval, OOV/noisy speech, long audio collections	[20] [21] [22] [23] [24] [25] [26]
Privacy requires explicit local-only design choices, not just local	To satisfy “no upload,” transcription, indexing, and search should run	Privacy-sensitive speech processing	[27] [28] [29] [30] [31] [32]

models	locally and avoid cloud APIs and telemetry		
Latency depends heavily on endpointing and token emission	User experience is shaped more by decoding behavior than by model size alone	On-device streaming ASR	[16] [33] [34]
Compression helps, but tradeoffs differ by bottleneck	Quantization can preserve WER with weak speed gains; distillation can speed up more but may hurt WER	Compressed ASR on edge/consumer-class devices	[35] [36] [37]
Noise, disfluency, and overlapping speech remain the main failure modes	Personal voice notes in real environments will reduce transcription and alignment quality	Spontaneous / noisy / conversational audio	[38] [39] [40] [41] [42] [43]

Direct Answer

Yes. The evidence supports building the feature you want as a **fully on-device pipeline**:

1. run **offline ASR** on voice files already stored on the laptop,
2. run **word-level alignment** to assign timestamps to each token,
3. build a **local search index** over the transcript and timing data, and
4. return a result that opens the right file and **jumps directly to the moment** where the searched phrase occurs [5] [6] [7] [9] [13] [15] [24].

For a practical implementation, the strongest evidence points to a **streaming RNN-T or Conformer-style first-pass recognizer**, a **word aligner such as WhisperX-like or NeMo Forced Aligner-style alignment**, and a **hybrid local retrieval stack**: exact transcript search first, phonetic or lattice-based fallback second, and optional semantic retrieval third for paraphrases or ASR misses [5] [6] [7] [13] [14] [15] [22] [24].

The main caution is that the supplied literature supports the components strongly, but does **not** clearly document a single published laptop product that proves the entire workflow end-to-end with explicit guarantees of **no upload, no telemetry, and clickable jump-to-word playback**. So the safest conclusion is: **the feature is feasible, but you will likely need to assemble it from proven local components rather than rely on one fully validated reference system** [16] [44] [45] [46] [47].

Confidence level: High — multiple source clusters independently support offline ASR, alignment, local retrieval, and privacy-by-local-processing, though end-to-end product validation is incomplete [1] [6] [7] [9].

Study Scope

This report covers research themes directly relevant to private laptop-based voice-note search: **on-device ASR architectures, word-level alignment, local transcript and phonetic retrieval, privacy-preserving offline design, laptop/edge performance tradeoffs, and user-facing jump-to-moment workflows** [9] [16] [17] [48] [49] [50]. The evidence spans classic foundational work and newer studies up to 2025, with emphasis on **offline operation, timestamp granularity, local indexing, and practical responsiveness** [18] [35] [51].

Assumptions & Limitations

- The evidence supports **component feasibility** more strongly than a single published end-to-end desktop system with all required guarantees [44] [45].
- Timestamping methods are promising, but the literature often lacks **standardized phrase-level timestamp error reporting** for long, imperfect, or personal recordings [52] [53] [54] [55].
- Laptop-specific benchmarks for **RAM, battery, and real-time factor** remain limited in the retrieved set [46] [56] [57].
- Real-world recordings with **noise, overlap, and disfluency** remain the main risk to exact phrase search quality [38] [39] [40] [43].

Suggested Further Research

- Benchmark a full local pipeline on consumer laptops using **WER, timestamp error, indexing time, search latency, RAM, and battery** under one protocol [46] [52] [57].
- Compare **transcript-only** retrieval against **hybrid transcript + lattice/phonetic + semantic** retrieval for personal voice-note collections [7] [24] [26].
- Evaluate phrase search on **short informal recordings** rather than benchmark speech corpora [4] [58].
- Study how **speaker overlap and diarization** affect jump-to-moment accuracy in voice notes [40] [47].
- Develop auditable methods to enforce **telemetry-free offline mode** in desktop applications [28] [31].

Full Report

Foundational Context

The requested feature combines four technical functions that must work together locally: **speech recognition**, **time alignment**, **search indexing**, and **playback localization** [2] [17] [49] [50]. Speech recognition converts audio into text; alignment maps text tokens back to exact time positions; retrieval finds candidate files and spans for a text query; and playback localization opens the file at the matched moment [6] [59].

For this use case, “private” does not just mean using a good model; it means the entire operational path—audio ingestion, transcription, indexing, and query execution—stays on the laptop without cloud upload or server-side inference [9] [10] [12] [32]. That requirement changes the engineering priorities: compact models, streaming or chunked decoding, local storage of transcripts and timing metadata, and local search structures become more important than maximizing raw benchmark accuracy alone [8] [16] [27] [58].

A text-like search bar over voice messages therefore needs two different kinds of precision: **lexical precision**, so the phrase is recognized or recoverable despite ASR errors, and **temporal precision**, so the result can jump to the right spoken instant rather than merely to the right file [17] [24] [59].

Confidence level: High — the core architecture is consistently supported across ASR, alignment, retrieval, and privacy literature [1] [5] [7] [9].

1. On-Device Speech-to-Text for Local Voice Search

Research on offline ASR for constrained hardware repeatedly emphasizes compact end-to-end architectures such as **RNN-T, MoChA, and chunked/transducer-style models** as the standard way to balance accuracy and latency on-device [16] [48] [58] [60]. More specifically, streaming **RNN-T/Conformer-style** systems are repeatedly presented as suitable for local or low-latency deployment [13] [16] [46] [61].

This matters because your tool needs to process voice files locally without waiting on a remote service. Streaming first-pass models are well suited to this because they are designed for incremental decoding and can outperform CTC-based baselines on both latency and accuracy in evaluated settings [13]. Chunked or streamable transformer decoding is also feasible when self-attention context is constrained, with only slight accuracy loss relative to full attention [14]. A practical pattern is a **fast first pass plus stronger second-pass rescoring**, using the first pass for responsiveness and the second for accuracy recovery [13] [15].

A key finding is that perceived responsiveness depends less on raw parameter count than on **token emission and endpointing behavior** [16]. Joint ASR plus endpointing improves the latency/WER tradeoff, and integrated endpointing can cut latency by more than 2x relative to using a separate endpointer without degrading quality [33]. For a desktop voice-note search app, this implies that fast “indexing complete” behavior will depend heavily on decoding strategy, not just model size [34].

Compression is relevant but not uniform in effect. Quantization can shrink model size with very small WER degradation, but may fail to improve speed much because of dequantization overhead [35]. Distillation can provide stronger speed and energy benefits, but at larger WER cost [35] [36]. Taken together, these findings suggest selecting compression according to the bottleneck: **RAM-limited systems may prefer quantization, while latency-limited systems may benefit more from distillation if the WER hit is acceptable** [35] [36].
Inferred: that deployment decision rule is a synthesis across reported tradeoffs rather than a single paper’s explicit prescription [35] [36].
Confidence level: High — architectural direction is strong; exact laptop performance remains underreported [56] [61].

2. Word-Level Timestamps and Phrase Alignment

Word-level timestamps are the essential bridge from “search found the phrase” to “jump to the moment it was spoken” [17] [59]. The evidence here is strong at the component level. **WhisperX** is reported to add voice activity detection and forced phoneme alignment to produce word-level timestamps for long-form audio, with state-of-the-art long-form transcription and word segmentation claims [5]. **NeMo Forced Aligner** outputs token-, word-, and segment-level alignments and is reported to have the best word-alignment accuracy and speed versus other aligners in the cited benchmark [6]. Iterative end-to-end timestamp estimation also outperforms CTC baselines and is comparable to GMM-HMM forced alignment [17].

For long or imperfect transcripts, the literature points to robust alternatives rather than assuming perfect verbatim text. Torchaudio’s long-form aligner is designed for approximate transcripts and can work with frame-wise CTC models [19]. Other workflows use phonetic decoding, anchor points, or fuzzy string matching to obtain useful timestamps even when the transcript is noisy or non-verbatim [55] [62] [63] [64] [65].

The main weakness is evaluation consistency. Several papers describe alignments as accurate or precise but do not report standardized word- or phrase-level timestamp error for the settings most relevant to personal recordings [52] [53] [54] [66]. The 2025 benchmark helps by showing neural aligners approach HMM accuracy on short utterances, while CTC segmentation is robust for long audio and improves with larger windows [18]. Still, the practical UX question—how often the jump lands exactly where a user expects in messy long-form speech—remains only partly quantified [52] [55].

Failure modes are clear. Alignment quality degrades around **reduced speech, pronunciation variability, fillers, repairs, and overlap** [38] [39]. Systems trained on cleaner corpora may silently drop disfluencies or normalize them into content words, which can damage both phrase search and timestamps [39]. Time-based segmentation that snaps results toward pauses or speech boundaries can improve search-and-jump behavior beyond raw word timing alone [67]. **Confidence level: High** for feasibility, **Medium** for exact UX quality in uncontrolled consumer recordings [5] [18] [38].

3. Local Indexing and Retrieval Over Voice Notes

Once transcripts and timestamps exist, the next challenge is search quality. Traditional transcript-only lookup is useful, but the literature repeatedly shows that **lattices, phonetic indexes, and two-stage retrieval** improve robustness when ASR is imperfect [24] [25] [26]. Early and modern spoken retrieval work converge on the same pattern: first use a compact index to generate candidates, then rescore or scan more precisely [20] [49].

For exact phrase search, a practical local strategy is:

1. **Search the 1-best transcript** for exact and near-exact phrase matches [68].
2. If no good hit appears, search a **phonetic or lattice-derived index** to recover OOV terms or ASR mistakes [22] [23] [24].
3. Optionally run **semantic retrieval** over transcript chunks for paraphrase-style queries [7] [8].

This hybrid design is well supported. Uniterm and related compact phonetic indices support fast coarse retrieval and outperform alternative lightweight methods in reported mobile voice-tagged experiments [21]. Phonetic-lattice systems on voicemail corpora matched in-vocabulary word-lattice baselines and stayed nearly as accurate for OOV keywords [22]. Position-aware lattice indexing improves ranking quality over 1-best indexing while also keeping the inverted index compact [24]. These findings strongly support keeping more than just the final transcript when building a search product [69] [70].

Semantic indexing is also relevant, especially when users do not remember the exact spoken wording. Offline transcript chunking plus embeddings in a local vector index supports real-time semantic retrieval over long spoken content [7]. Local vector stores can be used on-device without cloud services [8] [71]. However, semantic search complements phrase search rather than replacing it, because the user's requirement is explicit phrase lookup with jump-to-moment behavior [26].

Inferred: The most robust local product design is a **three-layer retrieval stack**: transcript phrase search first, phonetic/lattice fallback second, semantic search third [7] [24] [25] [72]. **Confidence level: High** — hybrid retrieval is supported across multiple retrieval traditions [20] [21] [22] [26].

4. Privacy-Preserving Fully Local System Design

The privacy requirement is not incidental; it is central. Multiple sources explicitly frame offline/local transcription as the privacy-preserving alternative to cloud ASR, particularly for sensitive recordings [1] [2] [27] [32]. Local processing reduces exposure to interception, third-party storage, and retention risks associated with cloud transmission [29] [30].

The evidence supports the feasibility of local-only stacks. Fully offline speech systems can use on-device ASR and browser-native or local capture/streaming without cloud services [9]. Offline assistants and local RAG systems show that ASR, retrieval, and response generation can operate without network access on edge devices or laptops [10] [11] [12] [73] [74]. Some studies explicitly enforce offline operation by running on secure local machines with no internet connection and using local Whisper, Kaldi, or offline scripts [28] [31].

The threat model also includes local data handling. Some workflows separate identifying metadata from transcripts or recordings and store participant data under random identifiers [30] [75]. For a voice-note search tool, that suggests separating:

- raw audio files,
- transcript/timestamp indexes, and
- user-visible metadata such as file names or conversation labels [30] [31].

The important caveat is that the literature does **not** clearly document one published desktop or browser app that proves the entire chain—capture, transcription, indexing, search, jump-to-moment—with explicit guarantees of no upload and no telemetry [44] [45]. So privacy assurance in a real product will depend on implementation discipline, not just model choice [28] [32].

Confidence level: High for the privacy rationale and local feasibility; **Medium** for auditable enforcement patterns in end-user desktop software [9] [44].

5. User-Facing Search and Jump-to-Moment Workflow

User-facing evidence is consistent with the desired interface. Spoken-content retrieval systems commonly map queries to **exact timestamps or clickable segments** using ASR transcripts and time-aligned metadata [59] [76]. An online audio indexing system explicitly timestamps ASR words so target segments can be rapidly identified and accessed [59]. Search-and-hyperlinking style systems segment long recordings so users can find precise relevant passages [77].

For long recordings, transcript-centric navigation matters. Prior work notes that returned audio can be minutes or hours long, so effective navigation depends on visualizations of transcripts and aligned ASR data [78]. Time-aligned transcripts give direct semantic access to audio, while browsing aids reduce the burden of finding the relevant passage manually [68] [79].

For your use case, the core workflow should be:

1. User enters a text query.
2. System searches local transcript and fallback indexes.
3. Result list shows file name, matched phrase, and local timestamp.
4. Clicking a result opens the audio and begins playback at the matched moment.
5. The transcript view highlights the matching words and nearby context [59] [68] [79].

A useful design refinement is to store both per-word timestamps and boundary information from pauses or segments, then snap playback to a natural speech onset rather than the literal first matched word [59] [67].
Inferred: this should improve perceived usability when timestamps are slightly noisy [67].
Confidence level: Medium-High — interface principles are well supported, but exact desktop UX comparisons are limited [78] [80].

6. Performance, Limitations, and Deployment Tradeoffs

The main deployment tradeoffs are **latency, memory, battery/energy, robustness, and recall under ASR error** [16] [36] [43]. Compression can help, but effects differ. Quantization greatly reduces size with minimal WER loss but may not materially improve speed [35]. Distillation can double speed and shrink the model more, but with noticeably worse WER [35]. Structured pruning and related efficiency methods can improve inference speed by around 3–4x in reported settings while exploring quality tradeoffs [81].

Noise robustness is a major concern for personal recordings. RNN-T is reported as more robust than MoChA to noise and reverberation in a streaming comparison [43]. Teacher-student learning and data augmentation can improve noisy-speech performance, sometimes with larger gains under noisy than clean conditions [41] [42]. Even so, uncontrolled voice notes remain difficult because transcription and alignment errors compound [40] [43].

There is also a notable evidence gap: the retrieved literature does not clearly provide a 2020–2025 offline laptop ASR system for voice-note workloads that reports both **WER and real-time factor on consumer CPUs/NPUs**, nor a unified benchmark covering **transcription, timestamp error, search latency, memory, and battery** [16] [46] [56] [57]. That weakens precise hardware planning, though not the broader feasibility conclusion [2] [9].

Confidence level: Medium — tradeoff directions are clear, but laptop-specific quantitative planning data are sparse [36] [57].

Conclusion

The literature supports a clear answer: **private, laptop-only voice-message search with word-level timestamps and jump-to-moment playback is technically achievable** by combining local ASR, local alignment, and local retrieval [2] [5] [7] [9]. The strongest architecture is a **streaming on-device recognizer**, followed by **word-level alignment**, then a **hybrid local index** that uses exact transcript search with phonetic/lattice fallback and optional semantic retrieval [6] [7] [13] [24].

The main implementation risk is not whether the parts exist—they do—but that the evidence base does not yet show a single published, fully validated desktop system with all the required guarantees and metrics in one place [44] [61]. So the research supports building the feature, while also implying that product quality will depend on careful systems integration, especially around timestamp quality, noisy recordings, and strict offline enforcement [16] [28] [38].

Overall confidence: High — strong convergence on feasibility and architecture; weaker support for one-stop end-to-end reference implementations [1] [6] [7] [44].

References

1. Sharma, A.S., Bhalley, R. (2016) ASR-A real-Time speech recognition on portable devices. *Proceedings - 2016 International Conference on Advances in Computing, Communication and Automation (Fall), ICACCA 2016*. https://doi.org/10.1109/ICACCAF.2016.7749004

2. Setiawan, P., Yusuf, R. (2022) IoT Device Control with Offline Automatic Speech Recognition on Edge Device. *12th International Conference on System Engineering and Technology, ICSET 2022 - Proceeding*. https://doi.org/10.1109/ICSET57543.2022.10010962

3. Ahmed, S., Roy-Chowdhury, A.R., Fawaz, K., Ramanathan, P. (2020) Prεεch: A system for privacy-preserving speech transcription. *Proceedings of the 29th USENIX Security Symposium*. https://www.scopus.com/pages/publications/85091905886

4. Shehu, O., Teliti, D., Kevric, J., (...), Karlik, B. (2025) Benchmarking Wav2Vec and Traditional Speech Recognition in Speech Transcription. *2025 6th International Conference on Communications, Information, Electronic and Energy Systems, CIEES 2025 - Conference Proceedings*. https://doi.org/10.1109/CIEES66347.2025.11300248

5. Bain, M., Huh, J., Han, T., Zisserman, A. (2023) WhisperX: Time-Accurate Speech Transcription of Long-Form Audio. *Proceedings of the Annual Conference of the International Speech Communication Association, INTERSPEECH*. https://doi.org/10.21437/Interspeech.2023-78

6. Rastorgueva, E., Lavrukhin, V., Ginsburg, B. (2023) NeMo Forced Aligner and its application to word alignment for subtitle generation. *Proceedings of the Annual Conference of the International Speech Communication Association, INTERSPEECH*. https://www.scopus.com/pages/publications/85171539894

7. Lin, I.-T., Chiu, C.-H., Liao, B.-T., Chen, B.-J. (2025) Integrating Large Language Models and Vector Databases for Podcast Semantic Search. *2025 IEEE 14th International Workshop on Computational Intelligence and Applications, IWCIA 2025 - Proceedings*. https://doi.org/10.1109/IWCIA68353.2025.11479780

8. Gu, X., Wang, Q., Pan, W., (...), Hu, R.Q. (2026) Structured Lightweight Speech Summarization with On-Device LLMs and Retrieval-Augmented Refinement. *2026 International Conference on Computing, Networking and Communications, ICNC 2026*. https://doi.org/10.1109/ICNC68183.2026.11416868

9. Di Leo, S., De Cicco, L., Mascolo, S. (2025) Real-Time Speech-to-Text on Edge: A Prototype System for Ultra-Low Latency Communication with AI-Powered NLP. *Information (Switzerland)*. https://doi.org/10.3390/info16080685

10. Supriya, A.V., Gowreesha, Dsouza, R.M., (...), Salian, S.N. (2025) A Privacy-Preserving Multimodal Voice Assistant with Offline Retrieval-Augmented Generation. *Proceedings of 2025 International Conference on Intelligent Systems and Pioneering Innovations in Robotics and Electric Mobility: Transforming Mobility and Automation Through Intelligent Engineering, INSPIRE 2025*. https://doi.org/10.1109/INSPIRE67328.2025.11300651

11. Kini, M., Jeevan, K., Ahazar, M., (...), Harshith, L.S. (2025) AI-Driven Desktop Voice Assistant. *Proceedings of International Conference on Sustainable Communication Networks and Application, ICSCN 2025*. https://doi.org/10.1109/ICSCN67106.2025.11308357

12. Suganya Devi, K., Sekar, K., Akshaya, V., (...), Surya Prakash, G. (2026) VESA: A Secure Offline Voice-Controlled Smart Home Ecosystem Using Agentic AI. *ADSSSC 2026 - International Conference on AI-Driven Solutions for Sustainable Smart Cities: Challenges and Opportunities*. https://doi.org/10.1109/ADSSSC67751.2026.11582655

13. He, Y., Sainath, T.N., Prabhavalkar, R., (...), Gruenstein, A. (2019) Streaming End-to-end Speech Recognition for Mobile Devices. *ICASSP, IEEE International Conference on Acoustics, Speech and Signal Processing - Proceedings*. https://doi.org/10.1109/ICASSP.2019.8682336

14. Zhang, Q., Lu, H., Sak, H., (...), Kumar, S. (2020) Transformer Transducer: A Streamable Speech Recognition Model with Transformer Encoders and RNN-T Loss. *ICASSP, IEEE International Conference on Acoustics, Speech and Signal Processing - Proceedings*. https://doi.org/10.1109/ICASSP40776.2020.9053896

15. Li, B., Chang, S.-Y., Sainath, T.N., (...), Wu, Y. (2020) Towards fast and accurate streaming end-to-end ASR. *ICASSP, IEEE International Conference on Acoustics, Speech and Signal Processing - Proceedings*. https://doi.org/10.1109/ICASSP40776.2020.9054715

16. Shangguan, Y., Prabhavalkar, R., Su, H., (...), Seltzer, M.L. (2021) Dissecting user-perceived latency of on-device E2E speech recognition. *Proceedings of the Annual Conference of the International Speech Communication Association, INTERSPEECH*. https://doi.org/10.21437/Interspeech.2021-1887

17. Yang, R., Cheng, G., Zhang, P., Yan, Y. (2022) An E2E-ASR-Based Iteratively-Trained Timestamp Estimator. *IEEE Signal Processing Letters*. https://doi.org/10.1109/LSP.2022.3190793

18. Deng, Y., Richardson, F., Steinberg, J., Torres-Carrasquillo, P. (2025) Speech-to-Text Forced Alignment Benchmark. *International Symposium on Image and Signal Processing and Analysis, ISPA*. https://doi.org/10.1109/ISPA66905.2025.11259471

19. Huang, R., Zhang, X., Ni, Z., (...), Metze, F. (2025) Long-Form Fuzzy Speech-to-Text Alignment for 1000+ Languages. *ASRU 2025 - 2025 IEEE Automatic Speech Recognition and Understanding Workshop*. https://doi.org/10.1109/ASRU65441.2025.11434643

20. Yu, P., Seide, F. (2005) Fast two-stage vocabulary-independent search in spontaneous speech. *ICASSP, IEEE International Conference on Acoustics, Speech and Signal Processing - Proceedings*. https://doi.org/10.1109/ICASSP.2005.1415155

21. Ma, M. (2008) Uniterm voice indexing and search for mobile device. *Proceedings - International Conference on Computer Communications and Networks, ICCCN*. https://doi.org/10.1109/ICCCN.2008.ECP.156

22. Seide, F., Yu, P., Ma, C., Chang, E. (2004) Vocabulary-independent search in spontaneous speech. *ICASSP, IEEE International Conference on Acoustics, Speech and Signal Processing - Proceedings*. https://www.scopus.com/pages/publications/4544257924

23. Siohan, O., Bacchiani, M. (2005) Fast vocabulary-independent audio search using path-based graph indexing. *9th European Conference on Speech Communication and Technology*. https://www.scopus.com/pages/publications/33745213806

24. Chelba, C., Acero, A. (2005) Position Specific Posterior Lattices for indexing speech. *ACL-05 - 43rd Annual Meeting of the Association for Computational Linguistics, Proceedings of the Conference*. https://doi.org/10.3115/1219840.1219895

25. Jones, G.J.F., Foote, J.T., Spärck Jones, K.Sparck, Young, S.J. (1996) Retrieving spoken documents by combining multiple index sources. *SIGIR Forum (ACM Special Interest Group on Information Retrieval)*. https://doi.org/10.1145/243199.243208

26. Saraçlar, M., Sproat, R. (2004) Lattice-based search for spoken utterance retrieval. *HLT-NAACL 2004 - Human Language Technology Conference of the North American Chapter of the Association for Computational Linguistics, Proceedings of the Main Conference*. https://www.scopus.com/pages/publications/85050187568

27. Hernández Acosta, L., Reinhardt, D. (2025) “Alexa, how do you protect my privacy?” A quantitative study of user preferences and requirements about smart speaker privacy settings. *Computers and Security*. https://doi.org/10.1016/j.cose.2024.104302

28. Dikaios, K., Rempel, S., Dumpala, S.H., (...), Uher, R. (2026) Sampling of natural speech for the assessment of psychopathology: data collection procedure and inter-rater reliability. *Psychiatry Research*. https://doi.org/10.1016/j.psychres.2025.116895

29. Lopez-Otero, P., Docio-Fernandez, L. (2021) Analysis of gender and identity issues in depression detection on de-identified speech. *Computer Speech and Language*. https://doi.org/10.1016/j.csl.2020.101118

30. Dutta, S., Irvin, D., Hansen, J.H.L. (2025) Exploring discrete speech units for privacy-preserving and efficient speech recognition for school-aged and preschool children. *International Journal of Human Computer Studies*. https://doi.org/10.1016/j.ijhcs.2025.103460

31. Henning, Zoe, Scholl, Isabelle, Hahlweg, Pola (2026) Psycho-oncologists' roles, tasks, and needs regarding requests for assisted suicide – a qualitative interview study on current experiences and future directions. *Supportive Care in Cancer*. https://doi.org/10.1007/s00520-026-10656-4

32. Haberl, A., Fleiß, J., Kowald, D., Thalmann, S. (2024) Take the aTrain. Introducing an interface for the Accessible Transcription of Interviews. *Journal of Behavioral and Experimental Finance*. https://doi.org/10.1016/j.jbef.2024.100891

33. Chang, S.-Y., Prabhavalkar, R., He, Y., (...), Simkó, G. (2019) Joint Endpointing and Decoding with End-to-end Models. *ICASSP, IEEE International Conference on Acoustics, Speech and Signal Processing - Proceedings*. https://doi.org/10.1109/ICASSP.2019.8683109

34. Shangguan, Y., Knister, K., He, Y., (...), Beaufays, F. (2020) Analyzing the quality and stability of a streaming end-to-end on-device speech recognizer. *Proceedings of the Annual Conference of the International Speech Communication Association, INTERSPEECH*. https://doi.org/10.21437/Interspeech.2020-1194

35. https://www.sciencedirect.com/science/article/pii/S1573772125000183

36. Peng, Z., Budhkar, A., Tuil, I., (...), Nassour, J. (2021) Shrinking Bigfoot: Reducing wav2vec 2.0 footprint. *SustaiNLP 2021 - 2nd Workshop on Simple and Efficient Natural Language Processing, Proceedings of SustaiNLP*. https://www.scopus.com/pages/publications/85134641134

37. Kacur, Juraj, Puterka, Boris, Pavlovicova, Jarmila, Oravec, Milos (2022) Frequency, Time, Representation and Modeling Aspects for Major Speech and Audio Processing Applications. *Sensors (Basel, Switzerland)*. https://doi.org/10.3390/s22166304

38. Yoon, T.-J. (2025) Robust forced alignment for Korean learners of English: A CrisperWhisper-MFA Comparison. *Proceedings of Meetings on Acoustics*. https://doi.org/10.1121/2.0002228

39. Muthu, M., Sasikala, E., Girirajan, S. (2026) Unified Architectures for Disfluency Detection and Transcription on Resource-Constrained Devices: A Systematic Review. *2026 International Conference on Data Science and Business Systems, ICDSBS 2026*. https://doi.org/10.1109/ICDSBS69077.2026.11634690

40. Mandal, A., van Hout, J., Tam, Y.-C., (...), Franco, H. (2013) Strategies for high accuracy keyword detection in noisy channels. *Proceedings of the Annual Conference of the International Speech Communication Association, INTERSPEECH*. https://www.scopus.com/pages/publications/84906224750

41. Mošner, L., Wu, M., Raju, A., (...), Hoffmeister, B. (2019) Improving Noise Robustness of Automatic Speech Recognition via Parallel Data and Teacher-student Learning. *ICASSP, IEEE International Conference on Acoustics, Speech and Signal Processing - Proceedings*. https://doi.org/10.1109/ICASSP.2019.8683422

42. Lakshminarayanan, V., Prud'Hommeaux, E. (2024) Exploring the impact of noise in low-resource ASR for Tamil. *DravidianLangTech 2024 - 4th Workshop on Speech, Vision, and Language Technologies for Dravidian Languages, Proceedings of the Workshop*. https://www.scopus.com/pages/publications/85189858669

43. Kim, J., Kumar, M., Gowda, D., (...), Kim, C. (2021) A Comparison of Streaming Models and Data Augmentation Methods for Robust Speech Recognition. *2021 IEEE Automatic Speech Recognition and Understanding Workshop, ASRU 2021 - Proceedings*. https://doi.org/10.1109/ASRU51503.2021.9688002

44. Reddy, S., Stanford, J.N. (2015) A web application for automated dialect analysis. *NAACL-HLT 2015 - 2015 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies - Demonstrations, Proceedings*. https://www.scopus.com/pages/publications/85039160898

45. Rosenberg, A., Audhkhasi, K., Sethy, A., (...), Picheny, M. (2017) End-to-end speech recognition and keyword search on low-resource languages. *ICASSP, IEEE International Conference on Acoustics, Speech and Signal Processing - Proceedings*. https://doi.org/10.1109/ICASSP.2017.7953164

46. Radfar, M., Barnwal, R., Swaminathan, R.V., (...), Mouchtaris, A. (2022) ConvRNN-T: Convolutional Augmented Recurrent Neural Network Transducers for Streaming Speech Recognition. *Proceedings of the Annual Conference of the International Speech Communication Association, INTERSPEECH*. https://doi.org/10.21437/Interspeech.2022-10844

47. Yandrapu, A., Lonka, S., Sreeram, A.S.K., Sunil Kumar, P. (2025) AUDIOSCOUT: Precise Audio Retrieval from Consecutive Interpretation Using Speaker Diarization and Natural Language Processing. *2025 3rd International Conference on IoT, Communication and Automation Technology, ICICAT 2025*. https://doi.org/10.1109/ICICAT68430.2025.11414632

48. Gonzales, M.G., Corcoran, P., Harte, N., Schukat, M. (2025) On-Device Evaluation of Joint Speech-Text Embeddings for ASR, TTS, and Speaker Recognition Tasks. *IEEE Access*. https://doi.org/10.1109/ACCESS.2025.3630503

49. Lee, H.-Y., Chou, P.-W., Lee, L.-S. (2014) Improved open-vocabulary spoken content retrieval with word and subword lattices using acoustic feature similarity. *Computer Speech and Language*. https://doi.org/10.1016/j.csl.2013.12.003

50. Lee, L.-S., Glass, J., Lee, H.-Y., Chan, C.-A. (2015) Spoken Content Retrieval - Beyond Cascading Speech Recognition with Text Retrieval. *IEEE Transactions on Audio, Speech and Language Processing*. https://doi.org/10.1109/TASLP.2015.2438543

51. Umesh, R., Sudharasan Dev, K., Praveen Kumar, M. (2025) AI-Powered Offline Voice Assistant for Rural Communities. *Proceedings of the 7th International Conference on Innovative Data Communication Technologies and Application, ICIDCA 2025*. https://doi.org/10.1109/ICIDCA66325.2025.11280419

52. Stan, A., Mamiya, Y., Yamagishi, J., (...), King, S. (2016) ALISA: An automatic lightly supervised speech segmentation and alignment tool. *Computer Speech and Language*. https://doi.org/10.1016/j.csl.2015.06.006

53. Martinc, Matej, Haider, Fasih, Pollak, Senja, Luz, Saturnino (2021) Temporal Integration of Text Transcripts and Acoustic Features for Alzheimer's Diagnosis Based on Spontaneous Speech. *Frontiers in Aging Neuroscience*. https://doi.org/10.3389/fnagi.2021.642647

54. Raccah, Omri, Chen, Phoebe, Gureckis, Todd M., (...), Vo, Vy A. (2024) The “Naturalistic Free Recall” dataset: four stories, hundreds of participants, and high-fidelity transcriptions. *Scientific data*. https://doi.org/10.1038/s41597-024-04082-6

55. Han, G., Murphy, J.G., Ladd, B.O., (...), Borsari, B. (2026) Leveraging Multimodal Self-Consistency Reasoning in Coding Motivational Interviewing for Alcohol Use Reduction. *Military Medicine*. https://doi.org/10.1093/milmed/usag224

56. Baquero-Arnal, P., Jorge, J., Giménez, A., (...), Juan, A. (2022) MLLP-VRAIN Spanish ASR Systems for the Albayzín-RTVE 2020 Speech-to-Text Challenge: Extension. *Applied Sciences (Switzerland)*. https://doi.org/10.3390/app12020804

57. Cohen, A., Rimon, I., Aflalo, E., Permuter, H.H. (2022) A study on data augmentation in voice anti-spoofing. *Speech Communication*. https://doi.org/10.1016/j.specom.2022.04.005

58. Kim, K., Jung, S., Lee, J., (...), Kim, D. (2019) Attention Based On-Device Streaming Speech Recognition with Large Speech Corpus. *2019 IEEE Automatic Speech Recognition and Understanding Workshop, ASRU 2019 - Proceedings*. https://doi.org/10.1109/ASRU46091.2019.9004027

59. Ajmera, J., McCowan, I., Bourlard, H. (2004) An online audio indexing system. *8th International Conference on Spoken Language Processing, ICSLP 2004*. https://www.scopus.com/pages/publications/85009071356

60. Fu, L., Li, S., Li, Q., (...), He, X. (2023) UFO2: A Unified Pre-Training Framework for Online and Offline Speech Recognition. *ICASSP, IEEE International Conference on Acoustics, Speech and Signal Processing - Proceedings*. https://doi.org/10.1109/ICASSP49357.2023.10096387

61. MacOskey, J., Strimel, G.P., Rastrow, A. (2021) Bifocal Neural Asr: Exploiting Keyword Spotting for Inference Optimization. *ICASSP, IEEE International Conference on Acoustics, Speech and Signal Processing - Proceedings*. https://doi.org/10.1109/ICASSP39728.2021.9414652

62. Boháč, M., Blavka, K. (2013) Text-to-speech alignment for imperfect transcriptions. *Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)*. https://doi.org/10.1007/978-3-642-40585-3_67

63. Bordel, G., Penagarikano, M., Rodríguez-Fuentes, L.J., (...), Varona, A. (2016) Probabilistic kernels for improved text-to-speech alignment in long audio tracks. *IEEE Signal Processing Letters*. https://doi.org/10.1109/LSP.2015.2505140

64. Hazen, T.J. (2006) Automatic alignment and error correction of human generated transcripts for long speech recordings. *INTERSPEECH 2006 and 9th International Conference on Spoken Language Processing, INTERSPEECH 2006 - ICSLP*. https://www.scopus.com/pages/publications/34547521678

65. Haubold, A., Kender, J.R. (2007) Alignment of speech to highly imperfect text transcriptions. *Proceedings of the 2007 IEEE International Conference on Multimedia and Expo, ICME 2007*. https://doi.org/10.1109/icme.2007.4284627

66. Meyer, T., Favaro, A., Oh, E.S., (...), Moro-Velázquez, L. (2025) Deep Stroop: Integrating eye tracking and speech processing to characterize people with neurodegenerative disorders while performing neuropsychological tests. *Computers in Biology and Medicine*. https://doi.org/10.1016/j.compbiomed.2024.109398

67. Eskevich, M., Jones, G.J.F. (2013) Time-based segmentation and use of jump-in points in DCU search runs at the search and hyperlinking task at MediaEval 2013. *CEUR Workshop Proceedings*. https://www.scopus.com/pages/publications/84924767326

68. Amir, A., Ashour, G., Srinivasan, S. (2004) Automatic generation of conference video proceedings. *Journal of Visual Communication and Image Representation*. https://doi.org/10.1016/j.jvcir.2004.04.008

69. Mangu, L., Brill, E., Stolçke, A. (2000) Finding consensus in speech recognition: Word error minimization and other applications of confusion networks. *Computer Speech and Language*. https://doi.org/10.1006/csla.2000.0152

70. Chia, T.K., Li, H., Ng, H.T. (2007) A statistical language modeling approach to lattice-based spoken document retrieval. *EMNLP-CoNLL 2007 - Proceedings of the 2007 Joint Conference on Empirical Methods in Natural Language Processing and Computational Natural Language Learning*. https://www.scopus.com/pages/publications/54049101376

71. Chakraborty, P., Ghosh, D., Srivastav, S., (...), Chakraborty, S. (2026) Privacy-First AI: Building an Offline RAG System with Streamlit and Local LLMs. *Lecture Notes in Electrical Engineering*. https://doi.org/10.1007/978-981-95-0429-9_3

72. Gavaldà, M., Schlueter, J. (2010) The truth is out there: Using advanced speech analytics to learn why customers call help-line desks and how effectively they are being served by the call center agent. *Advances in Speech Recognition: Mobile Environments, Call Centers and Clinics*. https://doi.org/10.1007/978-1-4419-5951-5_10

73. Bhardwaj, K., Hemalakshmi, G.R., Sharma, K., (...), Agrawal, A. (2025) An offline packaged RAG AI Assistant. *2025 International Conference on Decision Aid Sciences and Applications, DASA 2025*. https://doi.org/10.1109/DASA68193.2025.11498991

74. Tyndall, E., Wagner, T., Gayheart, C., (...), Langhals, B. (2025) Feasibility Evaluation of Secure Offline Large Language Models with Retrieval-Augmented Generation for CPU-Only Inference. *Information (Switzerland)*. https://doi.org/10.3390/info16090744

75. https://www.sciencedirect.com/science/article/pii/S2731455326001221

76. Varona, A., Nieto, S., Rodríguez-Fuentes, L.J., (...), Diez, M. (2011) A spoken document retrieval system for TV broadcast news in Spanish and Basque. *Procesamiento del Lenguaje Natural*. https://www.scopus.com/pages/publications/84855226768

77. Galuščáková, P., Pecina, P. (2014) Experiments with segmentation strategies for passage retrieval in audio-visual documents. *ICMR 2014 - Proceedings of the ACM International Conference on Multimedia Retrieval 2014*. https://doi.org/10.1145/2578726.2578753

78. Whittaker, S., Hirschberg, J. (2007) Accessing speech data using strategic fixation. *Computer Speech and Language*. https://doi.org/10.1016/j.csl.2006.06.004

79. Federico, M. (2000) System for the retrieval of Italian broadcast news. *Speech Communication*. https://doi.org/10.1016/S0167-6393(00)00022-4

80. Baume, C., Plumbley, M.D., Ćalić, J., Frohlich, D. (2018) A Contextual Study of Semantic Speech Editing in Radio Production. *International Journal of Human Computer Studies*. https://doi.org/10.1016/j.ijhcs.2018.03.006

81. Sang, J., Yolwas, Y. (2022) Compression Optimization Strategy for End-to-End ASR Model Based on Conformer. *Journal of Signal Processing*. https://doi.org/10.16798/j.issn.1003-0530.2022.12.018